

# Bhanu Pratap Sharma, PhD

Email: [bhanups1991@gmail.com](mailto:bhanups1991@gmail.com), [bhanupratap.sharma@univ-nantes.fr](mailto:bhanupratap.sharma@univ-nantes.fr)

Links: [Google Scholar](#), [LinkedIn](#), [Personal Webpage](#)

Phone: +33 0744 70 97 64

Researcher with combined industrial and academic experience in computational solid mechanics. Skilled in nonlinear stress analysis, impact mechanics, contact mechanics, constitutive modeling, experimental validation, and topology optimization.

## Core Skills

**Core FEA** : ABAQUS, NASTRAN, Altair HyperWorks, OptiStruct

**Programming** : Python, MATLAB, FORTRAN

**Other tools** : MS PowerPoint, MS Excel, MS Word, LaTeX, Linux

## Education

**Doctor of Philosophy** in Aerospace Engineering, IIT Bombay, India (2019 – 2025), CGPA 8.97/10

Thesis: **Computational Design of Foam-Based Shields for Blast Mitigation**

**Bachelor of Technology** in Mechanical Engineering, NIT Srinagar, India (2010-2014), CGPA 8.2/10

## Experience

**Research Engineer**, GeM Laboratoire, **Nantes University**, France (March 2026 – Present)

- Benchmarking of common methods (augmented Lagrangian, Nitsche) for solving contact problems with multiple solutions.
- Study of the influence of the parameters of these methods on convergence.

**PhD Research Scholar**, Aerospace Engineering, **IIT Bombay**, India (2019 – 2025)

- Implemented **custom VUMAT** and material parameter fitting viscoplastic PU foam.
- Analysed **metal and polymer foams** for **blast mitigation** using **finite element simulations**.
- Analysed **elastoplastic wave propagation** in metal and polymer foams subjected to shock loading.
- Collaborated on **topology optimization** of elastoplastic materials for energy absorption applications.
- Performed parametric studies via **surrogate modeling** using Python and MATLAB.
- Conducted experimental characterization of several collapsible materials using **shock tube**.

**Project Engineer**, Mechanical Engineering, **IIT Delhi**, India (Feb 2019 – May 2019)

- Reviewed literature and simulated trabecular bone response in LS-DYNA for anti-mine boot design.

**CAE Engineer**, Research and Development, **Maruti Suzuki**, Gurugram, India (July 2014 – May 2018)

- **Mesh generation** for complex powertrain geometries using **Altair Hypermesh**.
- **Nonlinear stress analysis** of exhaust cold end, engine mountings brackets, and fuel tank in **ABAQUS**.
- **Modal and FRF analysis** of fuel tank, engine oil sump, and exhaust cold end assembly in **NASTRAN**.
- Performed sequentially coupled **thermo-mechanical analysis** of full engine assembly in **ABAQUS**.
- **Topology optimization** of engine mounting brackets using **Altair Optistruct**.
- **Automation** of CAD pre-processing and mesh generation using **Altair SimLab** Process Automation.
- Followed **Kaizen philosophy** for **continuous improvement** and efficient simulation workflows.
- **Coordinated** with design and testing teams for **integrated design and development** decisions.